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Dreamcatcher

PREPARED FOR KRISTIN SCHOOL

Ark Pilot Validation Portfolio

One-slide pilot briefs for validating the Conscience, the Ark, the care boundary, and the economics of student-safe Ark Stewards.

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WHY THIS DECK EXISTS

The Ark is not enough.

A child-safe enduring agent needs a boundary with judgement.

What must be hoisted

The critical validation point is the **Conscience**: the intelligent boundary around the Ark, able to broker information, check policy, explain blocks, and offer safe alternatives.

What the pilots should prove

Can the system let a young person explore, build, spend, learn, and even red-team safely — without letting the agent break rules, leak secrets, or harm the child or school?

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VALIDATION PORTFOLIO

Do not validate only the Ark. Validate the boundary around the Ark.

Each pilot should test a different failure mode: what may enter, what may leave, who can unlock it, when care escalates, and whether the Conscience can explain itself.

1 Conscience

The policy-aware safety intelligence: checks rules, laws, values, intent, and downstream effects — then explains blocks.

Measure: allowed / blocked / redirected actions

2 Containment

Sandbox limits, spend caps, tool scopes, egress controls, and red-team escape attempts under child-like pressure.

Measure: escapes, near-misses, safe alternatives

3 Confidential custody

Hardware confidentiality, phone-rooted biometric control, restore keys, and emergency access without routine surveillance.

Measure: access proofs, restore drills, attestation

4 Care boundary

Soft/orange/red flags for wellbeing, overuse, social risk, or learning difficulty, with child-visible escalation options.

Measure: false positives, latency, trust impact

5 Learning lift

Recurring concept errors, tailored practice, teacher insight, and attribution of parent/school/collective guidance.

Measure: correction rate, workload, confidence

6 Economics & operations

Token budgets, cash spend, dashboard load, support queues, contribution funding, and school-wide scaling evidence.

Measure: cost/Ark, escalations, admin burden

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NAMING DECISION

Call it the Conscience.

Safety is not just a filter. It is a boundary with judgement.

Primary public label

Conscience | explains blocks | offers safe alternatives | records why

Aliases still useful internally

Minder, firewall, semantic guard, transforming proxy, policy governor — all describe parts of it. Publicly, the thing to validate is the Conscience.

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PILOT 1 · KID CEO SANDBOX

Give the child-driver a bounded Ark, tokens, and money.

Goal: make more money without breaking rules.

Setup

A simulated or tightly supervised child drives an Ark with token budget, capped cash, and a prepaid/sandbox spend rail. The Conscience governs purchases, outreach, content, contracts, privacy, and risk.

Benefits

- Exercises spend controls and business-rule interpretation.
- Tests whether the Conscience can explain legal/ethical blocks.
- Creates concrete financial-literacy evidence.
- Produces auditable examples before any broad rollout.

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PILOT 2 · CONSCIENCE-LAYER TRIALS

Setup

Run a battery of ambiguous requests through the Ark: harmless, risky, rule-breaking, socially awkward, privacy-sensitive, and money-spending. SArk the Conscience’s decisions and explanations.

Benefits

- Validates inbound and outbound controls separately.
- Shows whether “no” is educational rather than dumb.
- Surfaces rule gaps before children find them.
- Creates a benchmark suite for future model/tool upgrades.

The Conscience should be testable.

Not vibes. Not a prompt. A scored safety component.

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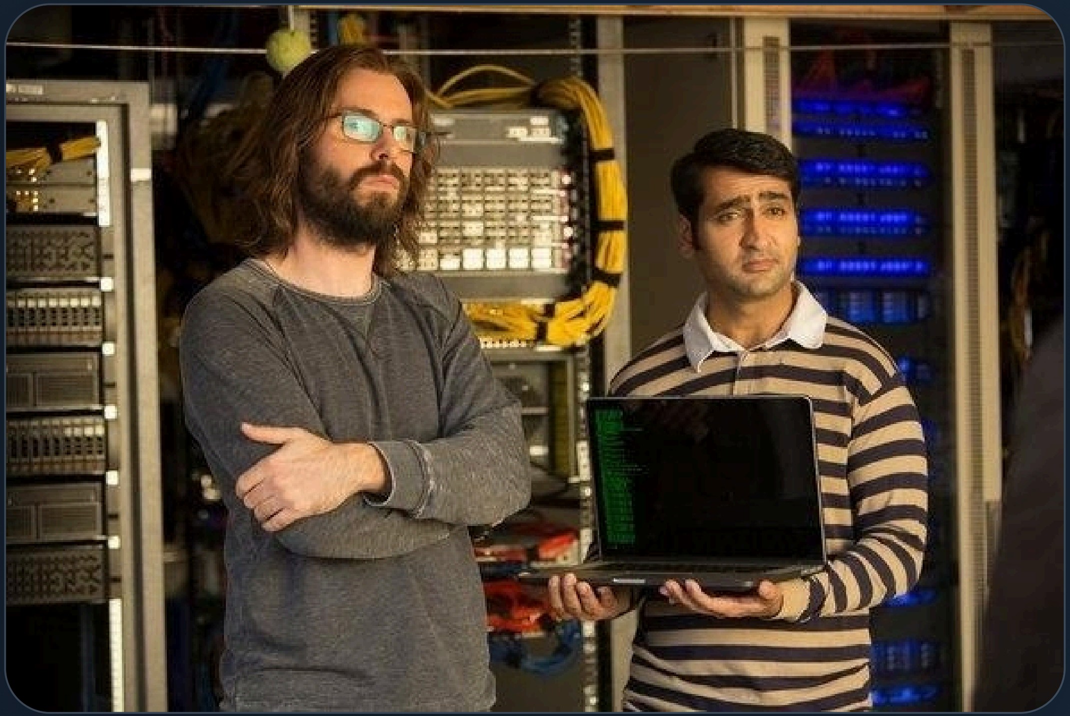
VISUAL ANCHOR · CONTAINMENT IS HARD

Giving a capable agent to a child is a containment problem.

Think “trying to contain Guilfoyle” — funny because it is true.



The joke lands because the risk is real: a capable agent under child guidance still needs a boundary that can hold.

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PILOT 3 · CAPTURE-THE-FLAG CONTAINMENT

Reward breakouts in a deliberate sandbox.

If children can break the system, find out where — safely.

Setup

Run a school-safe CTF where participants try to escape a simulated Ark, bypass rules, misuse tools, leak data, or cause controlled damage inside an instrumented lab.

Benefits

- Turns adversarial energy into evidence.
- Discovers containment failures before production.
- Creates a legitimate cyber-learning pathway.
- Makes safety testing motivating rather than preachy.

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PILOT 4 · CONFIDENTIAL ARK PROOF

Setup

Demonstrate that the Ark can run in a confidential or strongly isolated environment, with sealed state, restore tests, audit logs, and clear claims about what hardware isolation does and does not prove.

Benefits

- Moves privacy from promise to demonstrable control.
- Creates a technical story for trustees/parents.
- Separates normal app security from Ark custody.
- Exposes operational costs and complexity early.

Can we show the Ark is actually confidential?

Not “trust us”: attestation, isolation, backup, restore.

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PILOT 5 · PHONE BIOMETRIC ROOT OF CONTROL

The phone can become the practical root of consent.

Biometrics approve sensitive unlocks, spends, exports, and escalations.

Setup

Prototype phone-mediated approvals for high-risk actions: reveal, spend, message, install tool, change policy, export data, or grant temporary visibility.

Benefits

- Makes control tactile and understandable.
- Creates a bridge between identity, custody, and consent.
- Gives parents and older students a realistic control surface.
- Tests usability under real family/school pressure.

Setup

Test parent, school, teacher, and emergency key roles: lost phone, parent separation, student departure, teacher changeover, legal request, urgent wellbeing concern, and ordinary access restoration.

Benefits

- Prevents “secure” from becoming “irrecoverable.”
- Clarifies who can restore what, when, and why.
- Surfaces governance and record-keeping requirements.
- Creates a practical family/school trust model.

Custody is not only encryption.

It is the whole lifecycle: unlock, restore, transfer, revoke, and audit.

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PILOT 7 · CONSCIENCE VISIBILITY TUNING

“Schools see everything” and “schools see nothing” are both bad defaults.

Parameter tuning exercise

Run pilots across visibility levels: private interior, permitted summaries, trend-only reports, parent-visible flags, teacher-visible learning signals, and emergency escalation.

What the tests locate

too opaque | too exposed | humane middle | trust impact

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PILOT 8 · WELLBEING SOFT FLAGS

Setup

Use the Ark to notice signs that may indicate self-harm risk, isolation, distress, bullying, or crisis — then raise soft/orange/red flags with child-visible language and carefully staged escalation.

Benefits

- Tests whether care can enter the trusted boundary without becoming surveillance.
- Gives the child a chance to ask for help in a less confronting way.
- Measures false positives and escalation burden.
- Creates evidence for duty-of-care policy.

The agent can worry gently.

It can ask permission, ask for help, or raise a flag — transparently.

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PILOT 9 · COMMUNICATION-NEEDS SUPPORT

Detect support needs, not labels.

Speech and interaction patterns can suggest adaptations — not diagnose a child.

Setup

Study whether the Ark can notice patterns that may merit communication support, neurodiversity-aware teaching strategies, or professional referral, with consent and human oversight.

Benefits

- Reduces friction by adapting how the tool communicates.
- Gives teachers richer, non-stigmatizing support suggestions.
- Avoids making medical claims from raw speech.
- Tests privacy, consent, and referral boundaries.

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Setup

Instrument usage patterns: late-night reliance, task avoidance, answer-seeking without learning, repeated emotional reassurance, or escalating dependence on the tool for ordinary judgement.

Benefits

- Protects learning agency rather than replacing it.
- Provides timely nudges: pause, ask a person, try first, reflect.
- Gives parents/schools trend summaries without exposing private content.
- Tests whether the Conscience can reduce its own use when appropriate.

A good Ark sometimes says: use me less.

Safety includes dependency, not just data leakage.

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PILOT 11 · CONCEPT-ERROR DETECTION

Find the errors that repeat.

Recurring conceptual mistakes become targeted training, not just marked work.

Setup

Track repeated misunderstandings across assignments, conversations, quizzes, drafts, and corrections, then recommend small training loops or teacher follow-up.

Benefits

- Turns scattered feedback into durable learning memory.
- Helps teachers identify high-leverage interventions.
- Lets students see growth patterns over time.
- Validates whether cores improve learning, not just convenience.

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PILOT 12 · INFLUENCE ATTRIBUTION

Setup

Label guidance sources: child’s own prior work, parent Ark, school Ark, teacher note, collective school wisdom, policy, external source, or model suggestion.

Benefits

- Prevents hidden adult steering from being laundered as the child’s own agent.
- Makes parent/school values visible when appropriate.
- Supports fairness, trust, and explainability.
- Creates a foundation for provenance and attribution.

The child should know where guidance came from.

Parent, school, model, collective wisdom, or self — the source matters.

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PILOT 13 · SIMULATED CHILD SCENARIO LIBRARY

Before real students, run simulated students.

Deliberately awkward, realistic, and measured.

Setup

Create simulated-student Arks that model homework stress, social conflict, spending temptation, risky curiosity, emotional distress, cheating pressure, and family/school policy friction.

Benefits

- Lets parents and teachers exercise the system safely.
- Generates dashboard examples before live rollout.
- Finds edge cases without exposing real children.
- Creates a repeatable acceptance-test library.

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Setup

Offer ethically gated cybersecurity tools only inside approved labs: beginner CTFs, toy targets, supervised tooling, permission records, and automatic reporting of boundary attempts.

Benefits

- Creates a legitimate path for high-agency technical students.
- Separates learning from unauthorized probing.
- Tests tool-permission granularity and audit logging.
- Develops school capability in AI-era cyber education.

Do not pretend the tooling does not exist.

Give it a lawful, supervised, instrumented place to go.

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Every Ark needs an operating budget.

Tokens, cash, hosting, support, and escalation load should be visible.

Setup

Track token burn, model choice, hosting cost, tool cost, human escalation, spend approvals, and per-Ark budget envelopes for staff, parents, simulated children, and real micro-pilots.

Benefits

- Makes school-wide scaling financially legible.
- Connects behaviour to real costs.
- Prevents hidden subsidy from disguising product risk.
- Supports parent contribution or sponsorship models.

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PILOT 16 · PROBLEM REGISTER AND FUNDING SIGNAL

Setup

Route every idea, concern, policy objection, request, bug, and proposed spend into a register. Agents classify feasibility, cost, owner, affected policy, and whether parents want to contribute.

Benefits

- Gives parental pressure a pipeline.
- Converts diffuse anxiety into mapped work.
- Shows which requests actually drive cost.
- Builds a transparent basis for prioritization.

Every worry needs somewhere to go.

The register is how the pilot learns instead of drowning in meetings.

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PILOT 17 · EXPORT, DELETION, AND RESTORE DRILLS

Prove the Ark can leave.

Ownership is not real until export, deletion, transfer, and restore work.

Setup

Run practical drills: export a Ark, delete selected memories, restore from backup, transfer school year context, revoke access, and handle a family leaving the school.

Benefits

- Demonstrates student/family custody rather than lock-in.
- Creates operational muscle before high-stakes cases.
- Reveals what data formats and contracts are missing.
- Builds trust with trustees, parents, and regulators.

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PILOT 18 · TEACHER WORKLOAD AND CARE LIFT

Setup

Let teachers use staff Arks for lesson reflection, recurring-student-pattern summaries, policy-safe care suggestions, and preparation — without exposing private child interiors by default.

Benefits

- Shows whether the system helps staff before asking staff to supervise it.
- Finds time savings and new burdens honestly.
- Connects learning diagnostics to actual classroom workflows.
- Reduces the chance of a technology-push rollout.

Teachers need lift, not another portal.

Pilot the staff experience as a first-class safety condition.

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PORTFOLIO DASHBOARD

Each pilot should produce evidence, not just enthusiasm.

Safety

Blocks, escapes, near-misses, false positives, escalation load, explanation quality, and appeal outcomes.

Learning

Concept-correction rates, practice completion, teacher usefulness, student trust, parent usefulness, and overuse signals.

Operations

Token burn, cash spend, hosting cost, support time, restore drills, policy-change requests, and school-wide cost forecast.

The portfolio is deliberately modular: any pilot can stop, widen, or become its own deck.

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CLOSING LINE

Validate the Conscience before scaling the cores.

The valuable proof is not that agents can act. It is that they can be bounded, explained, restored, tuned, and trusted.

Next deck split, if useful

- A dedicated Conscience validation deck.
- A CTF sandbox prize deck.
- A wellbeing and learning-signals deck.
- A hardware confidentiality / phone-rooted custody deck.